



UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

Assignment-14

Author:

Name: MD. MIJU AHMED

ID: 2010676104

Supervisor:

SANGEETA BISWAS

ASSISTANT PROFESSOR

August 17, 2025

Contents

1	Introduction	2
2	Methodology	2
2.1	Dataset	2
2.2	Models	2
2.2.1	Fully Connected Feed Forward Neural Network (FCFNN)	2
2.2.2	Convolutional Neural Network (CNN)	2
2.2.3	Vision Transformer (ViT)	2
2.3	Training Setup	2
3	Results and Analysis	3
3.1	Comparison of FCFNN, CNN, and ViT	3
3.2	Effect of Number of Attention Heads	4
3.3	Effect of Patch Embedding Choice	5
3.4	Effect of Positional Embedding	6
4	Discussion	6
4.1	Challenges of ViT vs CNN and FCFNN	6
5	Conclusion	7

Abstract

This report investigates the performance of Vision Transformers (ViTs) in comparison with Convolutional Neural Networks (CNNs) and Fully Connected Feed-Forward Neural Networks (FCFNNs) on a 20-class subset of the ImageNet dataset. We explore how model architecture choices such as the number of attention heads, patch embedding methods, and positional embedding strategies influence ViT performance. The study highlights the advantages of ViTs over traditional models, while also discussing practical challenges in training and optimization.

1 Introduction

Image classification has been dominated by Convolutional Neural Networks (CNNs) due to their strong ability to capture spatial hierarchies. However, recent advancements in self-attention architectures, particularly the Vision Transformer (ViT), have challenged CNN supremacy. In this report, we train and evaluate ViTs, CNNs, and FCFNNs on a reduced ImageNet dataset consisting of 20 representative classes. We further study how architectural hyperparameters in ViT affect performance.

2 Methodology

2.1 Dataset

We used a 20-class subset of the ImageNet dataset, ensuring balanced class distribution. Images were resized to 64×64 for computational efficiency.

2.2 Models

2.2.1 Fully Connected Feed Forward Neural Network (FCFNN)

A simple FCFNN with 3 hidden layers was implemented. Each hidden layer used ReLU activation, and dropout was applied to reduce overfitting.

2.2.2 Convolutional Neural Network (CNN)

A CNN with 3 convolutional layers followed by max pooling and fully connected layers was trained. Standard ReLU activation and batch normalization were applied.

2.2.3 Vision Transformer (ViT)

The ViT model was trained with different hyperparameters:

- Number of self-attention heads: $\{2, 4, 6, 8\}$
- Patch embedding methods: linear projection, convolutional projection
- Positional embeddings: fixed sinusoidal vs learned embeddings

2.3 Training Setup

All models were trained for 30 epochs using the AdamW optimizer with a learning rate of $1e-3$ and weighted decay of $1e-4$. Cross-entropy loss was used. Data augmentation included random resized cropping and horizontal flipping. Here are uses 128 batch sizes for training and 256 batch sizes for validation.

3 Results and Analysis

3.1 Comparison of FCFNN, CNN, and ViT

Table 1 shows the test accuracy across three models.

Table 1: Model performance comparison on 20-class ImageNet subset

Model	Top-1 Accuracy (%)
FCFNN	21.10
CNN	55.60
ViT	44.00

Figure 1 shows the train and validation loss and accuracy curve.

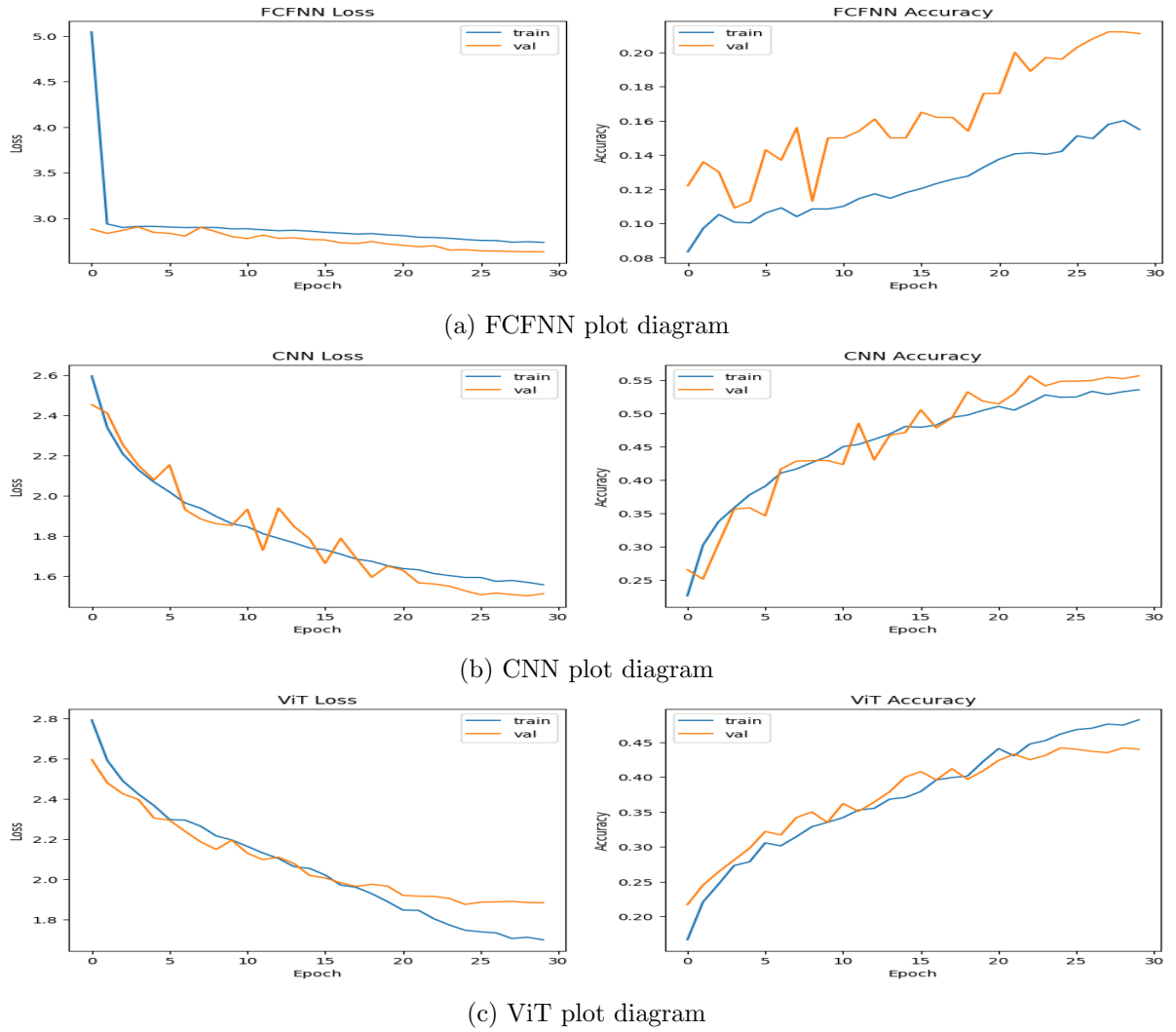
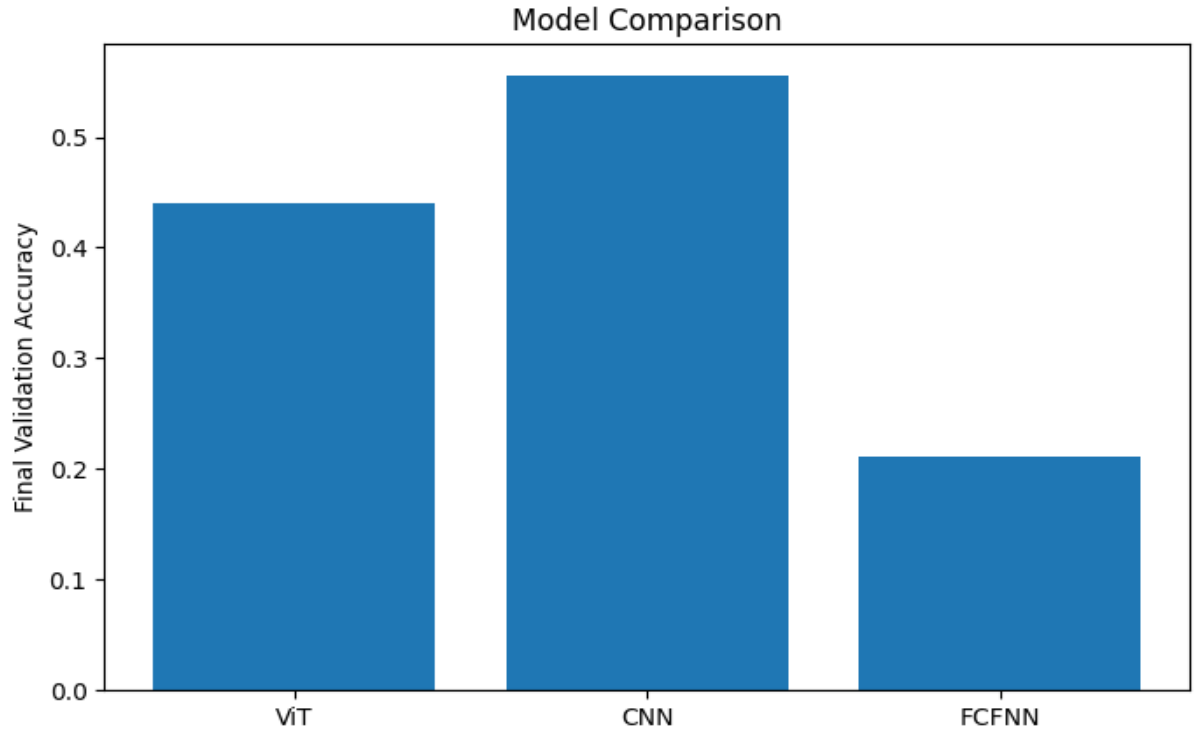


Figure 1: Plotted diagram for the FCNN, CNN and ViT models.

Figure 2 shows the accuracy comparison among FCNN, CNN and ViT models.



(a) FCNN plot diagram

Figure 2: Comparison bar for the FCNN, CNN and ViT models.

3.2 Effect of Number of Attention Heads

Figure 3 shows the performance of ViTs with different numbers of heads.

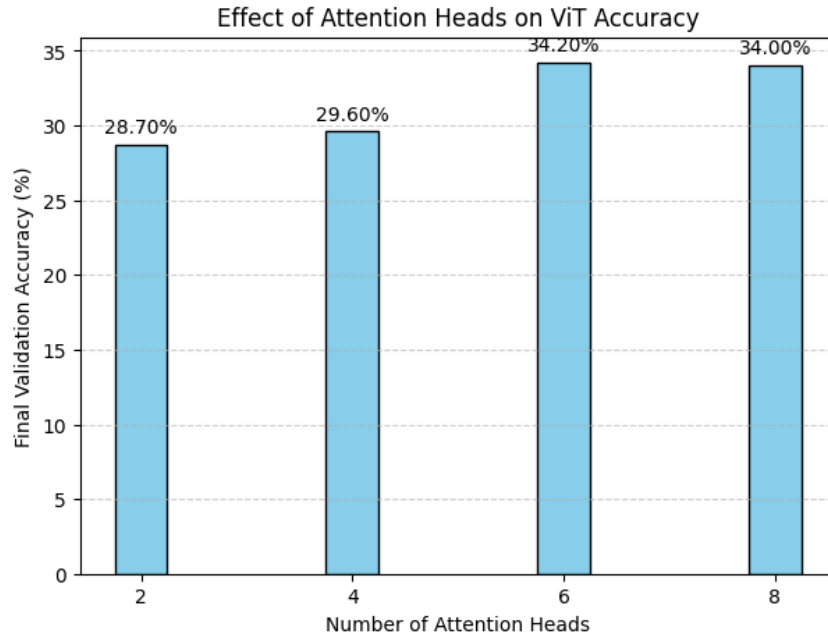


Figure 3: Effect of the number of attention heads on ViT performance.

Observation: After increasing the heads number, its increases the accuracy. After increasing the heads 6, its decreases little amount of accuracy.

Table 2: Effect of heads changes.

ViT-h	Accuracy (%)
ViT-h2	28.70
ViT-h4	29.60
ViT-h6	34.20
ViT-h8	34.00

3.3 Effect of Patch Embedding Choice

Table 3: Effect of patch embedding methods on ViT performance

Patch Embedding	Accuracy (%)
Linear Projection	34.20
Convolutional Projection	34.40

Observation: Convolutional embedding improved accuracy as it better preserves local spatial structure. Figure 4 shows the patch embedding performance.

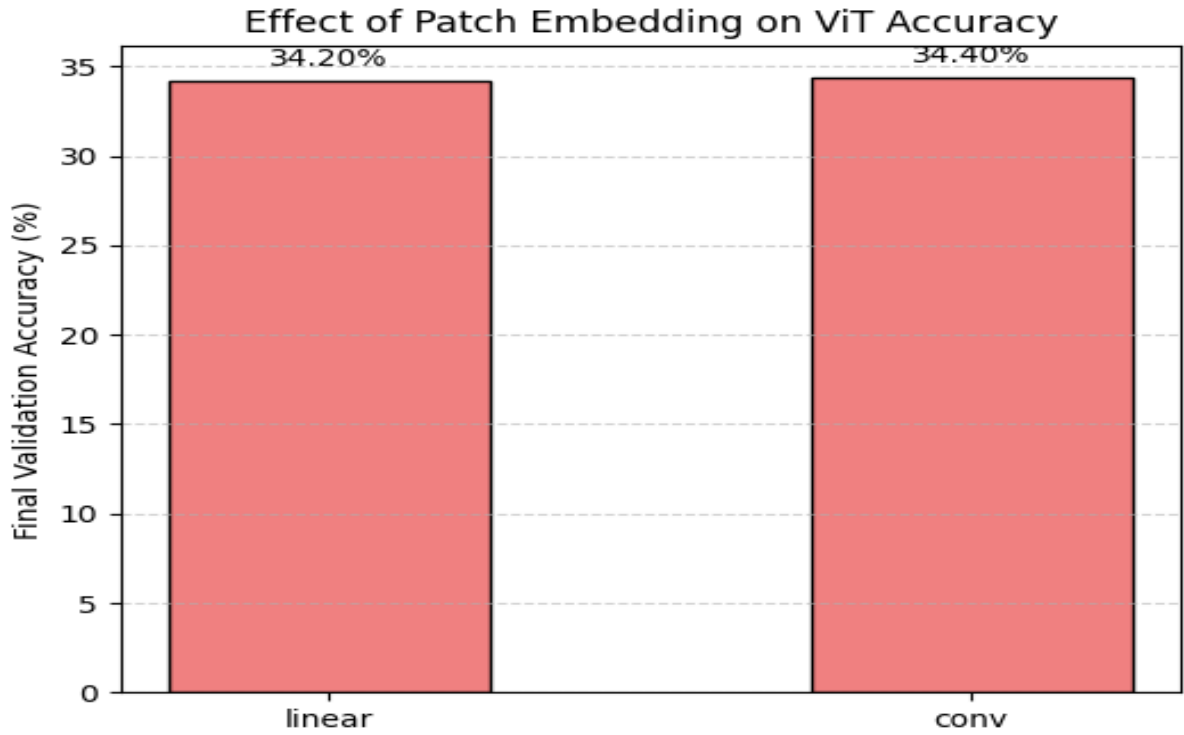


Figure 4: Comparison bar for patch embedding.

3.4 Effect of Positional Embedding

Table 4: Effect of positional embedding on ViT performance

Positional Embedding	Accuracy (%)
Fixed Sinusoidal	31.10
Learned	35.00

Observation: Learned embeddings yield better accuracy but sometimes suffer from instability early in training. Figure 5 shows the positional embedding performance.

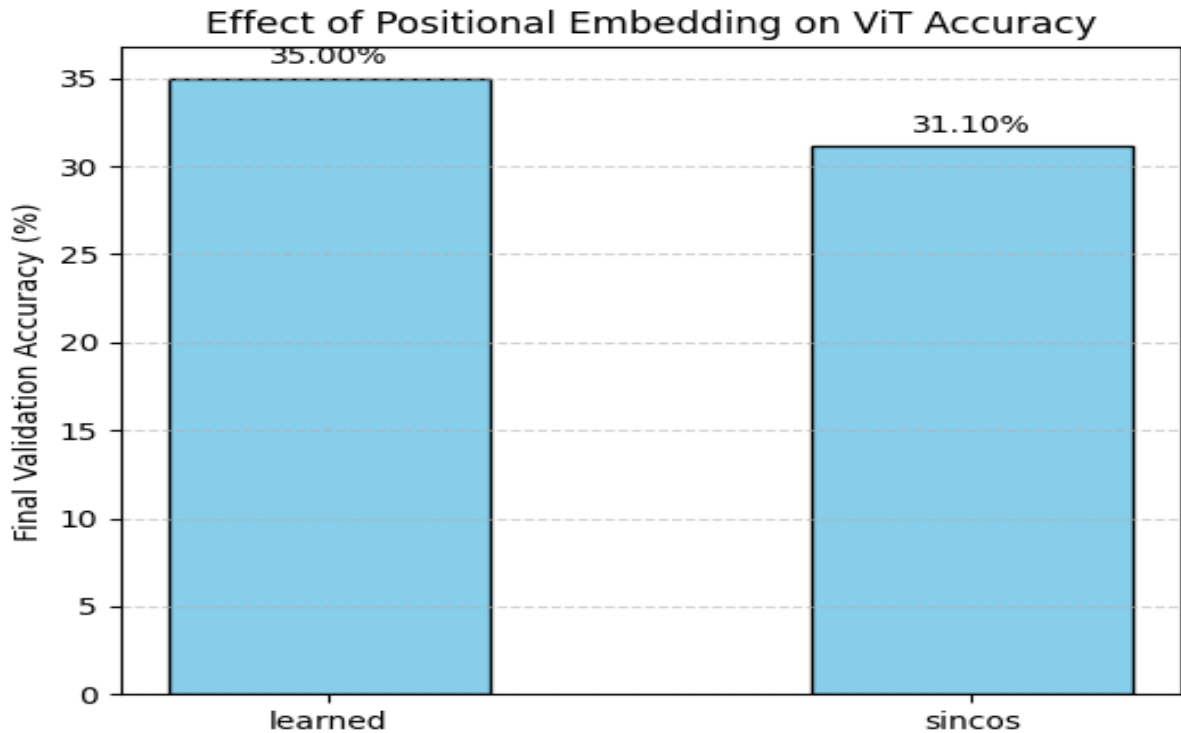


Figure 5: Comparison bar for positional embeddings.

4 Discussion

4.1 Challenges of ViT vs CNN and FCFNN

- **Data Requirements:** ViTs require larger datasets or heavy augmentation to perform well, unlike CNNs.
- **Training Complexity:** ViTs have higher training time and require careful learning rate tuning.
- **Model Size:** ViTs often have more parameters than CNNs, demanding higher memory.
- **Interpretability:** Self-attention provides better interpretability (attention maps) but complicates optimization.

5 Conclusion

This study confirms that Vision Transformers outperform traditional CNNs and FCFNNs on the ImageNet 20-class dataset. However, they introduce greater training complexity and resource requirements. Optimizing design choices such as attention heads, patch embeddings, and positional embeddings can significantly impact results. While CNNs remain efficient for smaller-scale problems, ViTs demonstrate strong generalization potential, particularly for large-scale vision tasks.

Assignment Github Link : Assignment